

Jay Patil

patjay0212@gmail.com | (510)-459-9157 | [LinkedIn](#) | [Profile](#) | [GitHub](#)

Summary

A passionate Full Stack Software Engineer with over two years of experience developing robust and scalable enterprise cloud web applications. Proficient in **Java, Python, MySQL, React and Angular**. With a knack for tackling complex challenges and strong problem-solving skills, my expertise spans areas like **streaming data pipelines, writing scalable microservices** and **strong understanding of cryptography**. Eager to drive innovation and learning new technologies.

Education

Rutgers University, New Jersey – *Master of Science in Computer Science*

Sept 2022 - May 2024

COURSES: Algorithms, Machine Learning, Distributed Systems, OS, DBMS

CGPA: 3.71/4.00

Pune Institute of Computer Technology, Pune – *Bachelor of Engineering in Computer Engineering*

Aug 2016 - June 2020

COURSES: Advanced Data Structures, Computer Networks, Computer Architecture

CGPA: 8.97/10.00

Experience

Antonina Mitrofanova Lab – Research Assistant, Software Developer

Sept 2022 – May 2024

- Designed and developed a **user-engaging and cross-browser compatible** web application for data analysis of complex human gene-splicing networks using HTML, CSS, JavaScript, Angular, and Cytoscape.js.
- Implemented **data feeds** from sources like NCBI ensuring seamless data integration and real-time updates.

Veritas Technologies LLC – Software Engineer, Full Stack

Sept 2020 – Jul 2022

- Worked in a **Scrum-based agile** environment with the Enterprise Vault.cloud (EVC) team to develop **intuitive user interfaces** to facilitate the data discovery from major data sources like Teams, Outlook, OneDrive, Bloomberg, etc.
- Took the ownership to update old code base to modern development standards making it more secure, **reducing loading time** of the application by **46%** using C#, HTML/CSS, TypeScript, Angular. Resulted in **big influx of new customers**.
- Implemented Azure REST APIs using C# to **enhance retrieval of data** from Azure Blob storage in .NET by 27%.
- Developed a scalable web service to integrate and connect Microsoft Teams services from Merge1 to EVC's internal platform.
- Coordinated efficient large-scale features deployments using Terraform and Puppet and managed CI/CD using TFS.

Projects

Text-to-Video Generation through Advanced Diffusion Models (GenAI)

- Spearheaded a pioneering text-to-video (T2V) generation project utilizing the "Tune-A-Video" methodology, focused on one-shot tuning of image diffusion models to elevate the quality and contextual accuracy of video output across diverse data domains. Utilized TensorFlow and PyTorch frameworks to meticulously implement and fine-tune the model.

Collaborative Editor

- Designed and developed a real-time collaborative editor that enabled multiple users to edit, collaborate and improve content simultaneously using Operational Transformation. Enhanced the system by integrating **Whisper** Voice to Text API to the system to support dictation. Implemented publish subscribe message broker functionality using Redis.

WeBuy – Online Auction System

- Developed easily scalable online auction system similar to eBay using NodeJS, React. Secured the system by implementing SSL/TLS for data transmission and database **encryption for sensitive user information**. Enhanced the system to handle multiple concurrent client requests and provide email alerts to users based on bid updates and new item availability.

Skills

Languages: Java, C#, Python3, JavaScript, SQL, C/C++, HTML/CSS, TypeScript

Web Technologies: NodeJS, ReactJS, ExpressJS, Angular, Flask, Web3, Google O'Auth

Tools and Frameworks: Git, Azure, Docker, MSSQL, MongoDB, JIRA, .NET, TFS, Elasticsearch, LangChain, AWS Redshift

Awards & Recognition

- Award for 'Team Collaboration and Inclusion' for collaboration with teams across the globe at Veritas.